pressure × 12



6-axis IMU

Stappone insoles
62.5 Hz · both feet
contact events · tags · GPX
4 walkers · 78 recordings

Step segmentation

vendor contact events
heel strike → heel strike
(one gait cycle, same foot)
toe-off → measured stance
≥ 5 samples

Step gate

drop transition / wait
drop multi-surface
≥ 80% one surface

Feature Extraction

- zone means at flat foot
- heel-curve shape & impulse
- loading rate, midstance
- CoP wander, spatial entropy

Feature Extraction

- impact peak at heel strike
- spectral centroid & entropy
- ZUPT-gated foot tilt
- step-to-step variability

raw step windows, per-channel z-scored

Intra-modality
Per-step segmentation & feature extraction

Feature vector

71 per-step features
− 21 person confounds
= 50 modelled

XGBoost

SVM, k-NN

per-classifier backward elimination · grouped leave-one-user-out CV

1D-CNN

- Surface class
  grass / loose / paved
- Compliance
  soft / hard
- Evenness score
  0-100
- Hardness score
  P(firm)

Cross-modality
Surface & compliance models